

Module 12: IEEE 802 LAN Standards & Ethernet (IEEE 802.3)

Unit 2: Data Link Layer & MAC Sublayer

IEEE 802 Architecture & Ethernet Evolution

LLC 802.2 vs MAC decomposition, complete IEEE 802.3 frame structure, formal mathematical derivation of the 64-byte minimum frame size, and 48-bit MAC addressing.

IEEE 802.3 Ethernet: Standard Frame Anatomy, Minimum 64-Byte Derivation & MAC Architecture

Standard IEEE 802.3 Ethernet Frame Format (Total Min: 64B, Max: 1518B)

Preamble 7 Bytes (1010...)	SFD 1B (10101011)	Dest MAC 6 Bytes (48 bits)	Src MAC 6 Bytes (48 bits)	Type/Len 2 Bytes	Payload Data (LLC PDU + Padding) 46 to 1500 Bytes (Requires Padding if < 46B)	FCS (CRC) 4 Bytes (CRC-32)
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Minimum Ethernet Frame = 6 + 6 + 2 + 46 + 4 = 64 Bytes (Excluding Preamble/SFD)

Maximum Frame Size = 6 + 6 + 2 + 1500 + 4 = 1518 Bytes (MTU = 1500 Bytes)

Interframe Gap (IFG) = 96 bit times (9.6 µs in 10 Mbps) mandatory between frames.

1. Why 64 Bytes Minimum Frame Size? (AKTU)

$T_t \geq 2 \cdot T_p$ (CSMA/CD Detection Law)
Sender must still be transmitting when collision noise returns!

In Standard 10 Mbps Ethernet (10Base5):

- Max network span = 2500 m + 4 repeaters.
- Max Round Trip Time ($2 \cdot T_p$) $\approx 51.2 \mu s$ (Slot Time).
- Bandwidth = 10 Mbps = 10^7 bits/s.

$L_{min} = 2 \cdot T_p \cdot B = 51.2 \mu s \times 10 \text{ Mbps}$
 $L_{min} = 512 \text{ bits} = \text{Exactly } 64 \text{ Bytes!}$

If payload < 46B, Padding (0s) is appended to reach 64B.

2. MAC Address Anatomy & Media Evolution

48-Bit MAC: 00:1A:2B : 3C:4D:5E
First 24 Bits (OUI): Assigned by IEEE to vendor (Cisco, Intel).
Last 24 Bits: Unique NIC Serial Number assigned by vendor.

Ethernet Standards Generations:

- Standard Ethernet (10 Mbps): 10Base5 (Thick), 10BaseT (Twisted pair).
- Fast Ethernet (100 Mbps, 802.3u): 100BaseTX (Cat5), Slot time = 5.12 µs.
- Gigabit Ethernet (1000 Mbps, 802.3z): Carrier Extension (512B slot).
- 10G Ethernet (10 Gbps, 802.3ae): Full Duplex switch only (NO CSMA/CD!).

Addressing Modes: Unicast (bit 0=0), Multicast (bit 0=1),
Broadcast: FF:FF:FF:FF:FF:FF (All 48 bits are 1s).

1. IEEE 802 Architecture & Frame Anatomy

The IEEE 802 architecture decouples transmission media specifics into the MAC sublayer while abstracting logical flow and network-layer multiplexing into the LLC 802.2 sublayer. The standard Ethernet frame spans between 64 and 1518 bytes.

2. 64-Byte Minimum Frame Size Derivation

Collision Detection Slot Time

Worst-case Round Trip Time ($2 \cdot T_p$) = $51.2 \mu\text{s}$

$L_{\min} = (2 \cdot T_p) \cdot \text{Bandwidth} = 51.2 \mu\text{s} \cdot 10 \text{ Mbps} = 512 \text{ bits} = 64 \text{ Bytes}$

Minimum Payload = $64\text{B} - 18\text{B} (\text{Header+Trailer}) = 46 \text{ Bytes}$